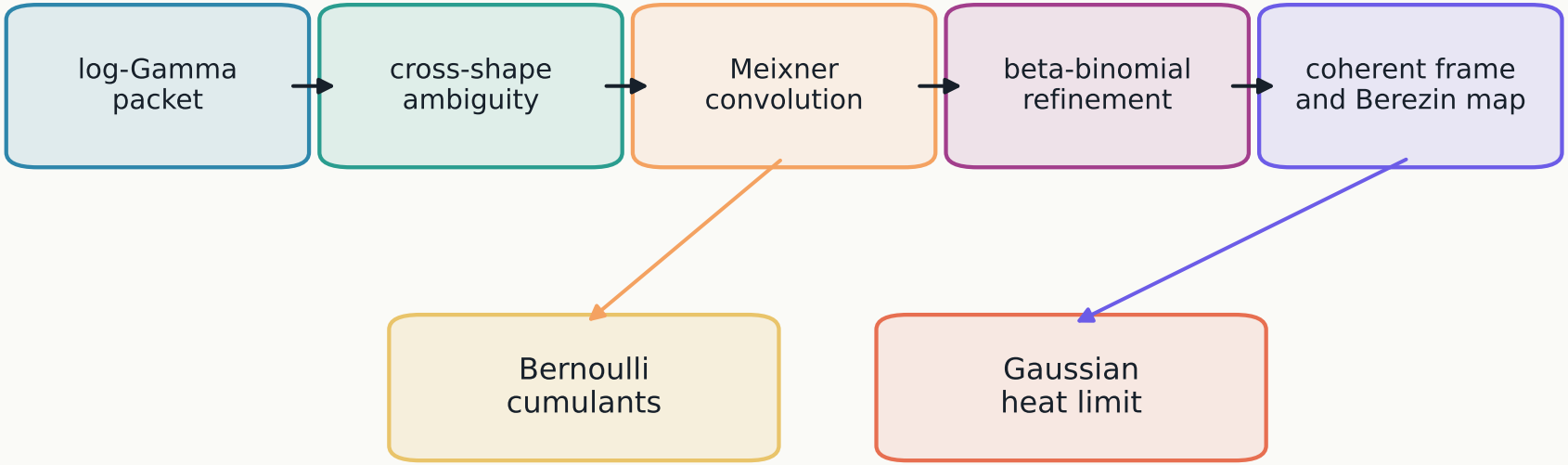From arithmetic shape to a packet semigroup